BERNSTEIN
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Global Aerospace & Defense

SpaceX, Howmet: Two ways to kill data centers on earth? Our Bernstein Data Center Day presentation



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Bernstein held its first Data Center Day this week, where we brought together Bernstein analysts from multiple sectors along with industry experts to discuss all aspects of the growing data center industry. As part of the day, there were seven analyst Best Ideas presentations. Here we provide our presentation, which focused on two investment ideas, SpaceX and Howmet. Each of these companies has the potential to limit terrestrial data center growth. We rate each Outperform.

SpaceX, with its IPO, is working to put a large constellation of data center satellites in space on sunsynchronous orbits. Depending on the growth of compute demand, this massive amount of new compute resource could limit the amount of terrestrial data center capacity needed. Terrestrial capacity will still be needed for very low latency work (e.g., high-frequency trading) and training. SpaceX's ability to place this compute capacity in space will depend on ramping up its launch program, particularly with fully reusable rockets. In addition, Starlink continues to be a great business with high margins. But, it will not ultimately be the business that can drive a multi-trillion dollar valuation. That business should be orbital data centers.

Howmet, with its leadership position in high pressure turbine blades, can constrain the industrial gas turbine market. Without blades, there will not be IGT growth and data center growth would be significantly limited. Howmet has more than a 50% share in IGT blades, just as it does in aero engine blades. These positions have come through investments in automation and production processes over the last several years that has created a competitive advantage that is difficult to break. The end result is that Howmet is able to capture lucrative long-term contracts both for IGT and aero applications, leading to strong margin growth. For IGT, the company has new contracts with seven major IGT providers for capacity expansion, which should begin coming online in Q4. We see the recent decline in the stock price over concerns about the commercial aero aftermarket and the GE/CPP deal as a buying opportunity.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our Global Research Disclosure Website.

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BERNSTEIN TICKER TABLE

			8 Oct 2026		TTM	Adjusted EPS				Reported P/E (x)	
			Closing	Price	Rel.						
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
HWM (Howmet)	O	USD	222.56	298.00	1.1%	USD	3.77	5.28	6.33	60.0	40.5	35.2
SPCX (SpaceX)	O	USD	160.57	248.00	NA	USD		(0.19)	1.37		(845.0)	117.4
SPX			7,765.36									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

SPCX valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate SpaceX Outperform, with a target of $248. We rate Howmet Outperform, with a target of $298.

DETAILS

TWO COMPANIES - TWO WAYS TO KILL DATA CENTERS ON EARTH?

I am going to be the bad guy at this Data Center Day Best Ideas panel. We have to best ideas and they each have potential limit the success of data centers on earth:

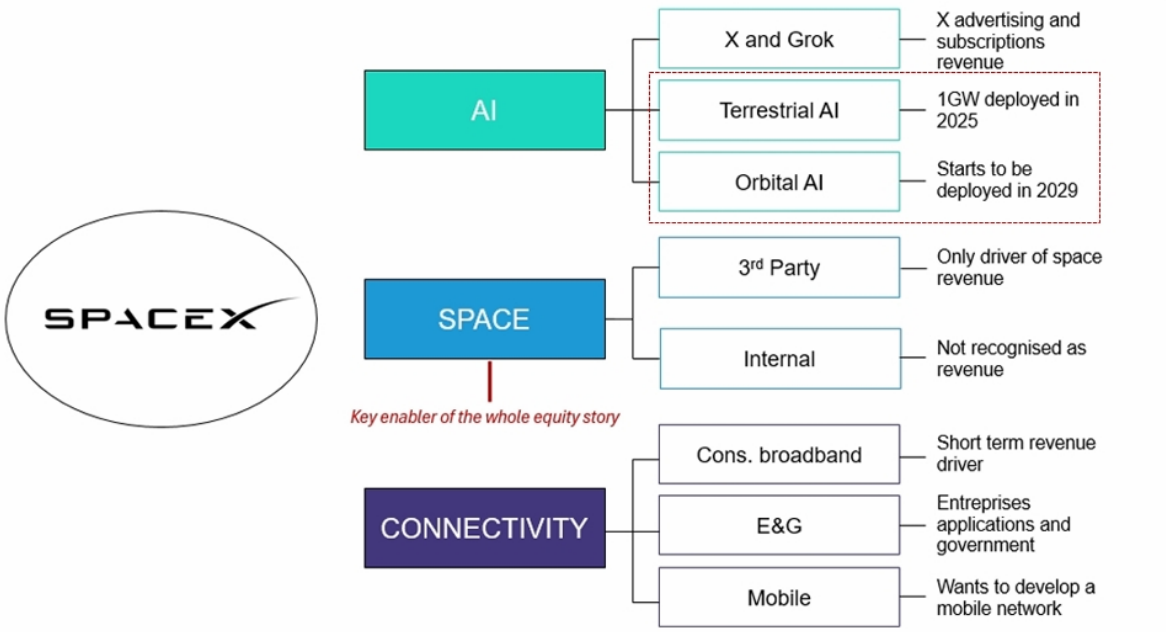
1. **SpaceX:** The core of SpaceX value is the prospect of placing massive amounts of compute capacity into orbital data centers. If SpaceX is successful, long-term capacity requirements on earth could be reduced. It is not as though ALL compute capacity would move to space. For very low latency applications and training, we expect terrestrial dat b a centers will remain dominant. But, as inference becomes the biggest driver of compute demand, the space alternative becomes highly attractive. The attractiveness comes from the elimination of many costs, but more importantly, from the ability to grow capacity rapidly, assuming that launch capacity can ramp quickly. See also, "*Space Exploration Technologies Corporation: Doing Time in the Universal Mind - Initiate at Outperform; Target Price $239*" and "*SpaceX: Upside from AI pricing, as Starship progress exceeds capex worries; Outperform, raising target to $248*".

2. **Howmet:** Back on earth, much of data center development requires power, with industrial gas turbines (IGTs) as the leading near-term option. Because IGTs depend on turbine blades as a critical component, the fact that Howmet has the lead share in an industry with view options, places it in an unusually powerful position. This is the same position that Howmet has in the aero engine industry. In both cases, Howmet largely holds the cards and is able to negotiate favorable contracts that lead to high margins. See also, "*Howmet: Airfoil competition controversy? Yes - but, expect continued strength; Q3 preview*".

SPACEX: MOVING DATA CENTERS TO SPACE

SpaceX is comprised of three segments (Exhibit 1), with the origin of SpaceX coming from its Space segment in 2002. While the business has always been tied to Elon Musk's multiplanetary vision, including colonization of Mars, that is far away, with a lot to happen in the interim. The source for a multi-trillion dollar valuation for SpaceX will be the AI business, and particularly orbital data centers. But, the key to unlocking the value in AI will be the space launch capability, where rapid reuse of rockets has the potential to place thousands of satellites in orbital. No competitor is remotely close to being able to provide that launch capacity of SpaceX. The third segment, Connectivity, is a great business with high growth and high margins. Many of you have probably used Starlink, at least on airline flights, where the performance is as good as broadband at home. But, it will require more than Starlink to support our $248 target price.

EXHIBIT 1: **Data centers will be SPCX key growth driver in the coming years**

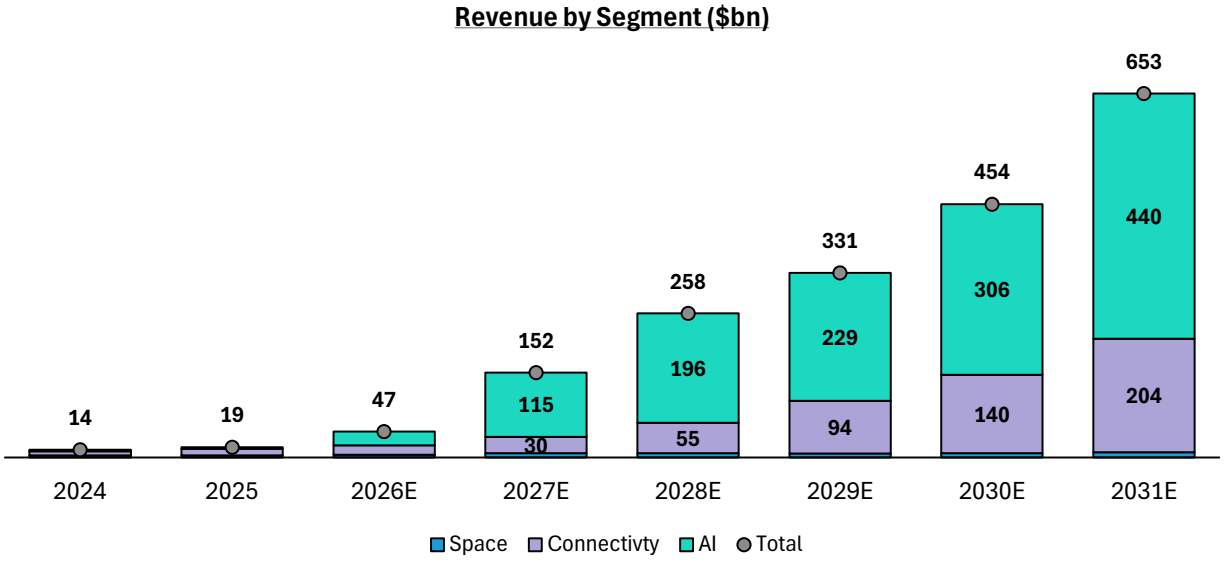


Source: Company reports, Bernstein analysis

Revenue to be led by AI

We are forecasting that the company will generate $653bn in revenue in 2031, with the main source of revenues coming from the AI business (Exhibit 2). We are well below the SpaceX outlook, which is now for $1 trillion in revenues in 2030. But, you do not need to go all the way to SpaceX's estimates to make the stock work. The primary challenge, in our view, will be getting the launch cadence up. That launch cadence depends primarily on reaching rapid reusability of its Starship rocket.

EXHIBIT 2: **Expect AI to dominate SpaceX revenues**



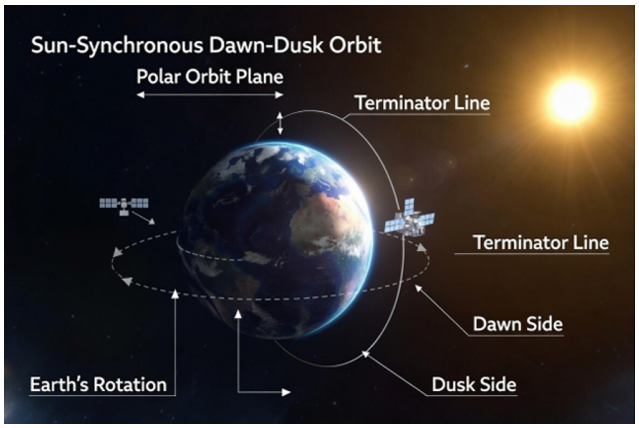
Source: Company reports, Bernstein analysis and estimates

We believe orbital data centers should be technically feasible

SpaceX's orbital data center looks technically feasible. Power comes from near-continuous solar generation in orbit (Exhibit 3). Cooling is solvable through radiative heat rejection. Laser inter-satellite links are already proven at scale on Starlink, which now has more than 11,000 satellites in orbit. Latency to Earth is not a constraint for most AI workloads, especially as demand moves from training to inference. The real hurdle is launch: the key variable is launch cost and cadence, which Starship's reusability is meant to address.

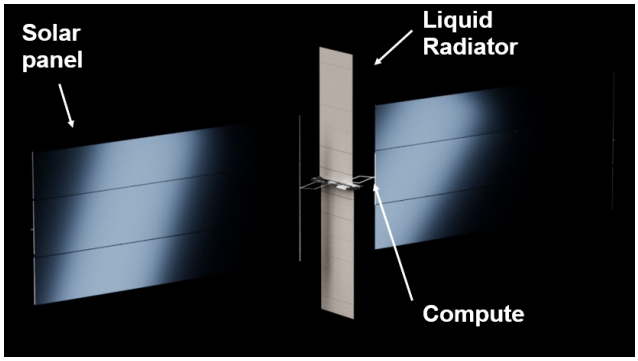
Each orbital data center satellite is a large deployable structure. It stands about 30m tall when deployed, has a 75m wingspan, and carries a Vera Rubin rack as its compute payload (Exhibit 4). The payload delivers 250 kW at peak and 175 kW on average. Roughly 30 satellites fit on each Starship launch, which adds up to more than 5 MW of compute capacity per launch.

EXHIBIT 3: **A sun-synchronous dawn-dusk orbit hugs the terminator line, keeping the satellite in the sun**



Source: Grok

EXHIBIT 4: **SpaceX orbital datacenter, Starmind**



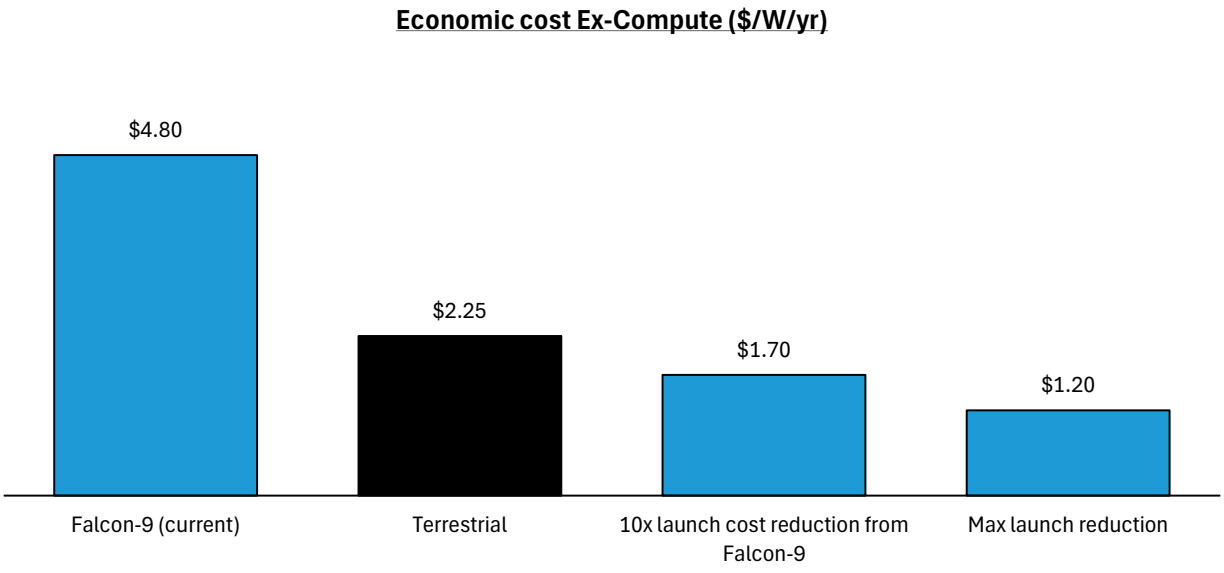
Source: Company image, Bernstein analysis

Orbital data center costs should be attractive - But, availability will likely be more important

Orbital data centers could cut total cost by 25% (excluding the cost of the rack) or more compared with terrestrial facilities (Exhibit 5). In orbit, operators avoid grid power costs, because the satellite runs on solar energy. They also avoid dedicated cooling systems and the land, buildings and other facility infrastructure a ground site needs. Orbital sites also skip permitting delays and local opposition (NIMBY), which now slow many terrestrial data center projects.

We believe that simple availability will likely be more important than the cost differential. For terrestrial data centers, there is opposition now in most locales. The permitting process is now very slow and the availability of craft labor for construction is also becoming a bottleneck. IGT production may also become a bottleneck (e.g., the issues in the next section on Howmet). Going to orbital data centers eliminates these bottlenecks. If launches can be done rapidly, the timeline for buildout of compute capacity could be much faster.

EXHIBIT 5: **Orbital data centers could cut total cost by 25% or more compared with terrestrial facilities**

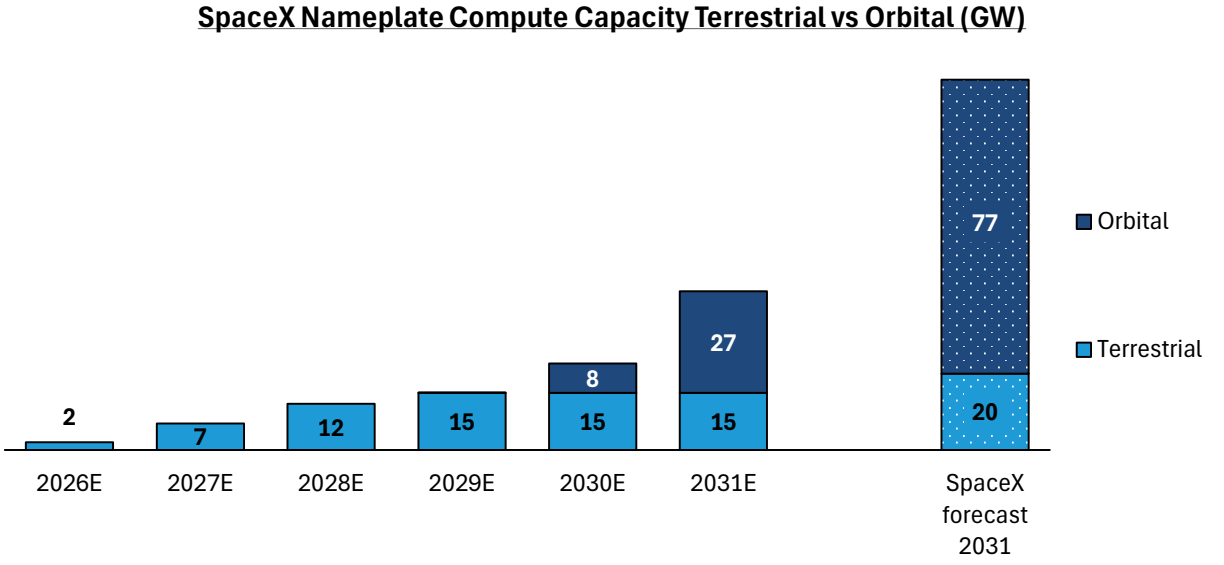


Source: Company reports, Bernstein analysis and estimates

Our orbital data center estimates are conservative relative to those of SpaceX

We model 27GW of orbital data center capacity by 2031 (see Exhibit 6), well below SpaceX's forecast of 77GW. The gap comes mainly from launch cadence, which we expect to ramp more slowly than SpaceX assumes. We also expect the first significant orbital capacity in 2030, a year later than SpaceX's 2029 target.

EXHIBIT 6: **We model 27GW of orbital data center capacity by 2031, well below SpaceX's forecast of 77GW**



Source: Company reports, Bernstein analysis and estimates

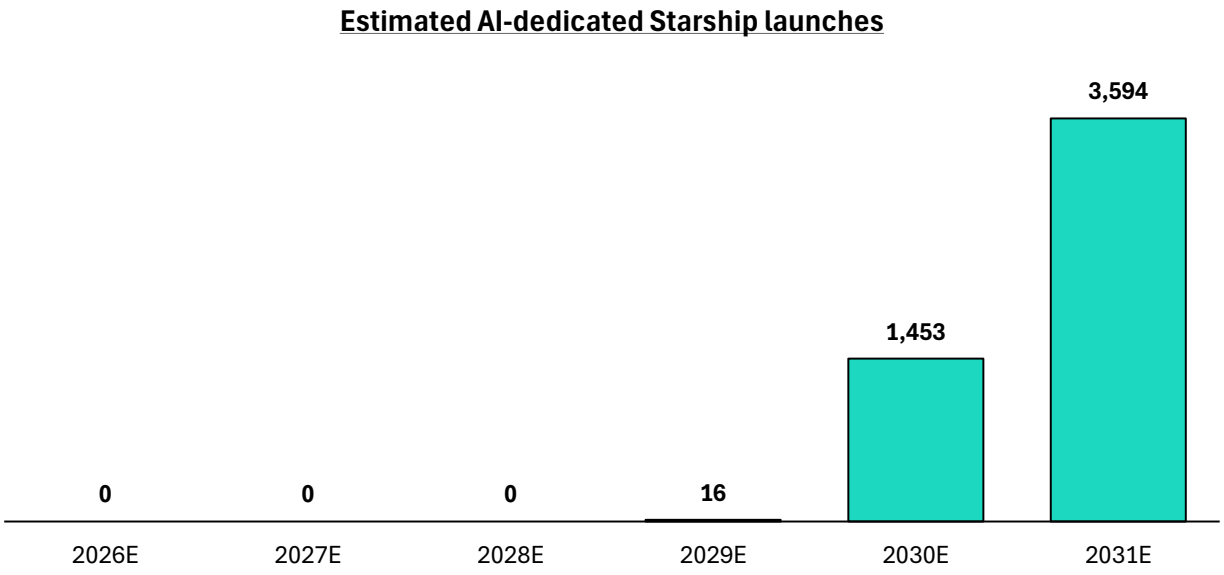
We expect nearly 4,000 Starship launches by 2031 - No competitors can come close

Our model assumes nearly 4,000 Starship launches in 2031, covering AI and other missions. Of those launches, more than 3,500 would be for data centers (Exhibit 7). This number of launches is far below the SpaceX forecast, which we believe would

require nearly 10,000 launches in 2031 to meet its objectives.

Starship has completed 14 orbital launches so far. SpaceX has demonstrated a booster catch but not a second-stage catch, which could come as early as launch 15. Heat shield performance is the critical test for reuse, and reuse underpins the whole economic case. The heat sheild appears to have held up well on launch14. But, we await the more detailed company assessment.

EXHIBIT 7: **By 2031, we estimate 3,594 AI-dedicated launches**



Source: Company reports, Bernstein analysis and estimates

Valuation: A $248 price target

We value SpaceX on 2031 estimates, using a 12m forward EV/EBITDA multiple (Exhibit 8). For the AI business, we set the multiple at the median of a basket of AI-proximate stocks. AI drives most of the valuation, but Connectivity (Starlink) remains a strong business in its own right. We discount it at 35%, which represents a high-end level for late stage venture financing.

Choosing comparable is difficult. For example, a neocloud model on earth cannot have near the rate of growth as SpaceX orbital AI, assuming that a rapid launch cadence is succesful.

EXHIBIT 8: **Orbital datacenters drives most of the company's valuation**

	12m forward EV/EBITDA	2031 EBITDA	SOP valuation	Discount rates	Discounted valuation
AI	21	367,694	7,574,487	35.0%	**2,856,067**
Connectivity	15	113,623	1,681,625	25.0%	**814,276**
Space	13	4,556	59,224	10.0%	**43,448**
				Total	**3,245,273**
				Target price	**248**

Source: Company reports, Bloomberg, Bernstein analysis and estimates

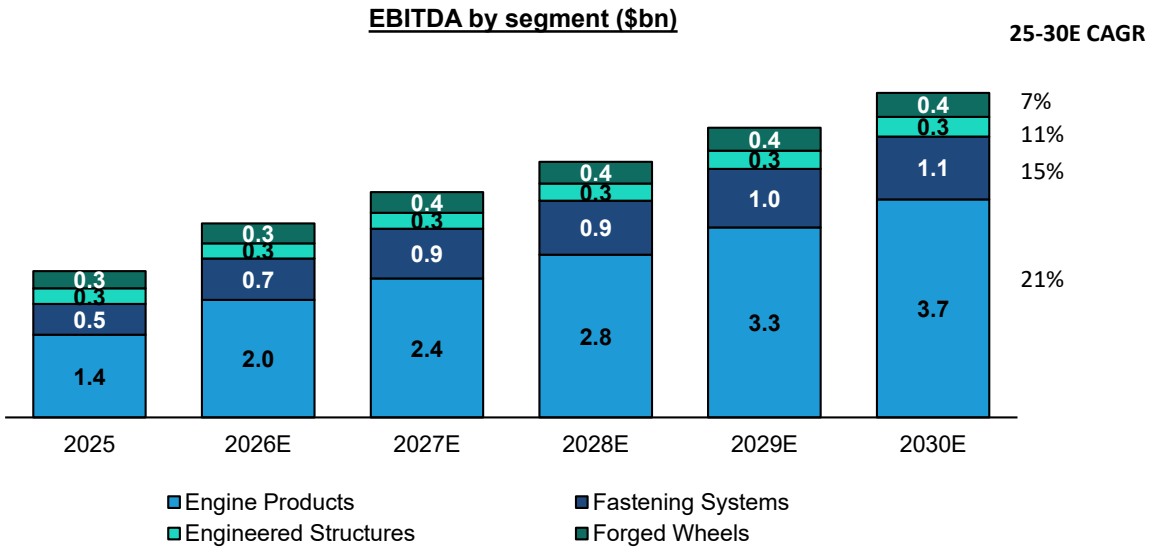
HOWMET: NO BLADES, NO IGT, NO TERRESTRIAL DATA CENTERS

Howmet is the leading producer of airfoils for aero engines and IGTs. The company's strong position in Engine Products, plus good performance by its other business led the stock to rise 500% over three years heading into August. But, during the last two months, the stock has pulled back by 20%, as have other aero engine companies. That pullback was exacerbated by concerns around SpaceX planning to build its own turbine blades and GE's planned acquisition of CPP. We see the company's exposure to the legacy aero aftermarket as fairly low. The SpaceX decision should be irrelevant and we see the GE/CPP deal as a relatively minor threat (see, "*Howmet: Airfoil competition controversy? Yes - but, expect continued strength; Q3 preview*").

Engine Products grow fastest

Engine Products is Howmet's highest-growth segment. We forecast an 18% revenue CAGR and a 21% earnings CAGR (Exhibit 9). The other business units are also strong. Engineered Structures is the exception only because divestitures hold its growth down.

EXHIBIT 9: **Engine Products is Howmet's highest-growth segment**



Source: Company reports, Bernstein analysis and estimates

Airfoils are the bottleneck in IGT

Turbine airfoils are the bottleneck, as the quotes in Exhibit 10 lay out how well-known this in the turbine markets. Howmet leads in manufacturing technology and yields. As a result, it has been able to set prices in both aero and IGT markets as part of long-term contracts. Because of its position, Howmet can ensure that its contracts limit risk exposure, should the market weaken. It is not investing ahead of demand, which can make it a constraint on growth, as Elon Musk has acknowledged.

EXHIBIT 10: **Blades are the bottleneck in the turbine industry**

> *"It's the vanes and blades in the turbines that are the limiting factor"*
>
> *"There are only three casting companies in the world that make these, and they're massively backlogged."*
>
> Elon Musk on X

> *"There is one company that can raise prices whenever it wants (on aero engine HPT blades) and there is nothing we can do about it"*
>
> Aircraft engine executive

Source: X, interviews

Contracts with all of the large IGT players

Howmet now has long-term supply agreements with seven major IGT providers and holds the majority share of airfoils at six of the seven largest. It is adding capacity in Japan, Europe and the US, though new capacity takes more than two years to bring online. IGT customers are already revisiting and raising their demand outlooks, mainly because of data-center power needs. CEO John Plant has said the IGT outlook will likely be raised in the coming months. That outlook is for revenue to double to $2bn within 3–5 years from 2025. Our estimates are already higher at a 25% CAGR.

EXHIBIT 11: **Howmet now has long-term supply agreements with seven major IGT provider...**



Source: Company reports, Bernstein analysis

EXHIBIT 12: **... and holds the majority share of airfoils at six of the seven largest**



Source: Company reports, Bernstein analysis

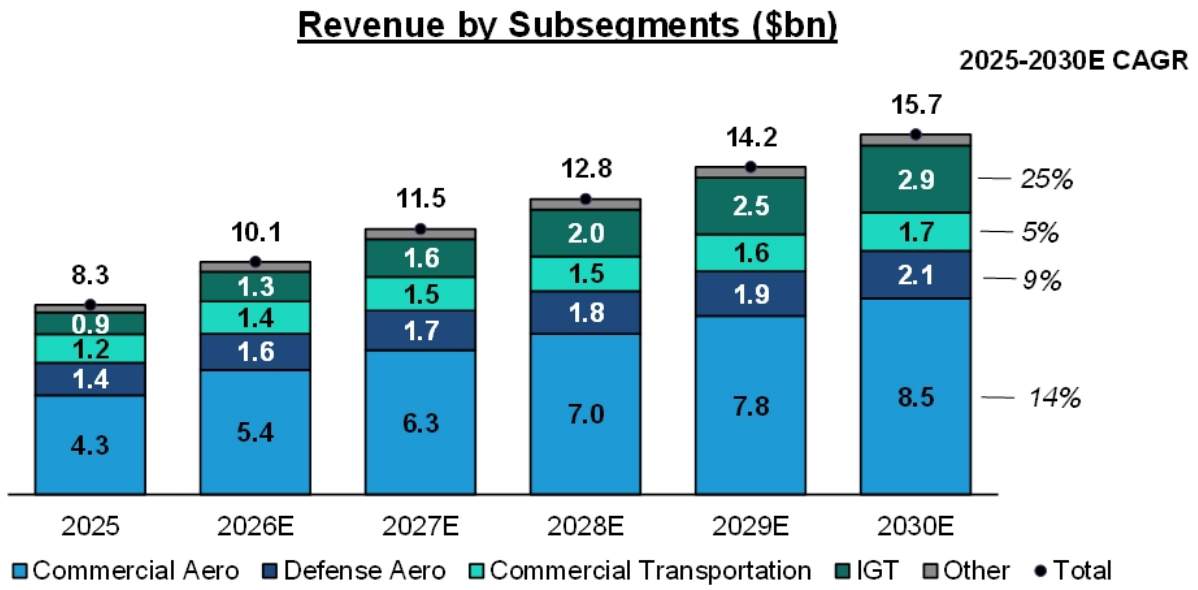
IGT is fastest growing subsegment in Engine Products

As Exhibit 13 shows, the largest subsegment in Howmet Engine Products is commercial aero. But, the fastest growing subsegment is IGT.

Howmet holds more than 50% of the market for IGT blades. IGT revenue grew 38% in 2Q26, which shows how strongly demand for data-center power is feeding through to its results. The new capacity increases have not even come online yet.

Capacity in Japan should be online in Q4.

EXHIBIT 13: **We expect very strong growth for the IGT subsegment, driven by demand from data-centers**

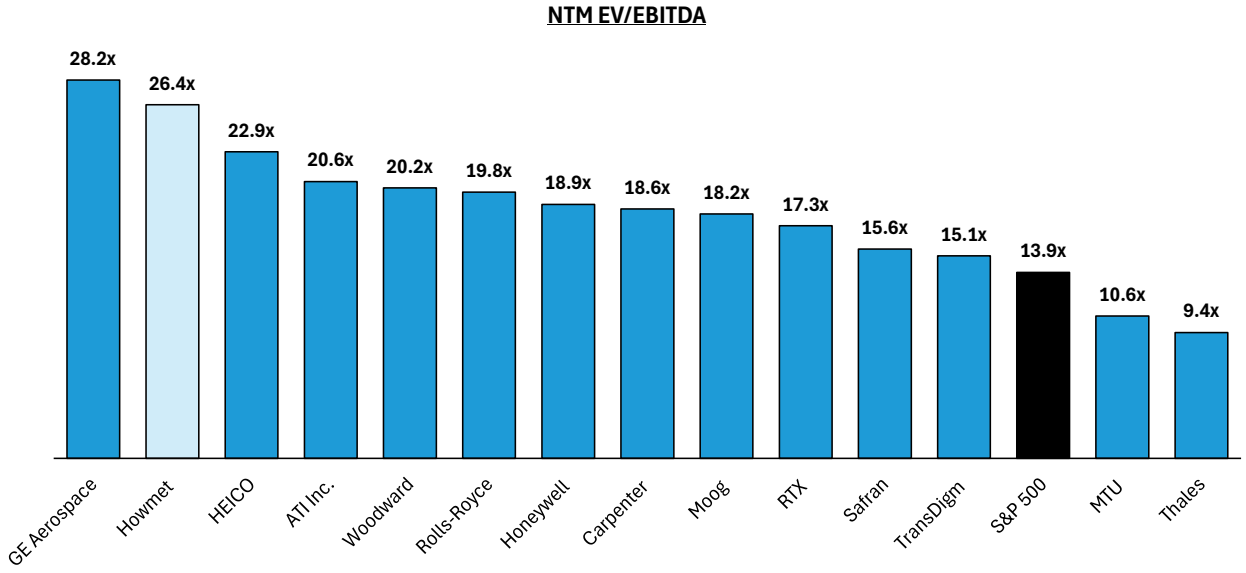


Source: Company reports, Bernstein analysis and estimates

Outperform - Target $298

We rate Howmet Outperform with a $298 price target. The stock is expensive, but we think it is worth the price. It trades at 26.9x NTM EV/EBITDA, the second-highest in our group after GE Aerospace at 28.3x. Both multiples have come down recently with concerns over high fuel prices. We see this as a good time to buy the stock.

EXHIBIT 14: **We rate Howmet Outperform with a $298 price target**



Source: Bloomberg, Bernstein analysis and estimates

I. REQUIRED DISCLOSURES

References to "Bernstein" or the “Firm” in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK（サンフォード・C・バーンスタイン株式会社）and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY

Howmet Aerospace Inc

For Howmet, we use a relative EV/EBITDA multiple of 180% on a 16.1x market multiple, which is roughly consistent with the middle of a range of peers. We apply the multiple to earnings four years forward and discount the valuation back to reach our 1-year price target of $298, adding discounted interim cash to shareholders.

Space Exploration Technologies Corporation

We base our valuation approach on 12-month forward EV/EBITDA applied to 2031 EBITDA. We believe it is necessary to use a long-term period (i.e., 2031) because the cornerstone of the value is in the orbital data centers. Applying a multiple to, say, 2028 EBITDA would not capture any of that business - i.e., it would simply be space launch, terrestrial compute and near-term Starlink. We use a sum of the parts methodology to give an EV/EBITDA valuation in 2030 (based on 2031 values). We then discount that valuation backward for each segment to mid-2027 in order to reach a one year target price of $248.

RISKS

Howmet Aerospace Inc

Risks to our price target for Howmet particularly relate to potential weakness in commercial aircraft demand as well as industry supply chain issues that could constrain growth. There can also be execution risks to margins as Howmet works to expand capacity to meet growing demand.

Space Exploration Technologies Corporation

Downside Risks to our Outperform rating and $248 price target include the ability to dramatically growth Starship launch capacity over time, which depends on full reusability. SpaceX must be able to operationalize its orbital data centers to deliver on the valuation. Semiconductor availability could also be a constraint, as could other potential supply chain issues. There are also risks associated with potential launch incidents or regulatory restrictions. Should compute demand be more limited than most expectations, that would form another risk.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

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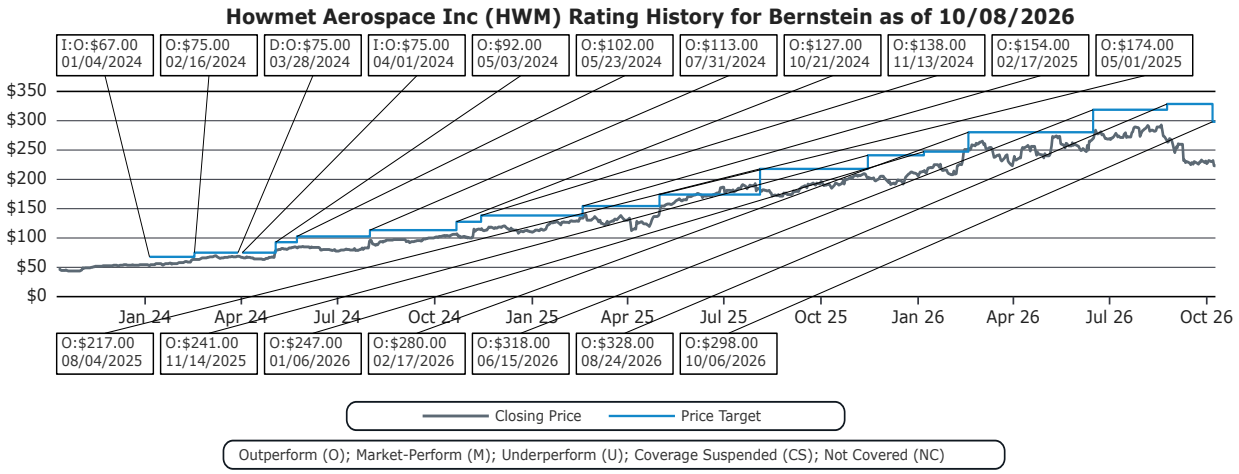
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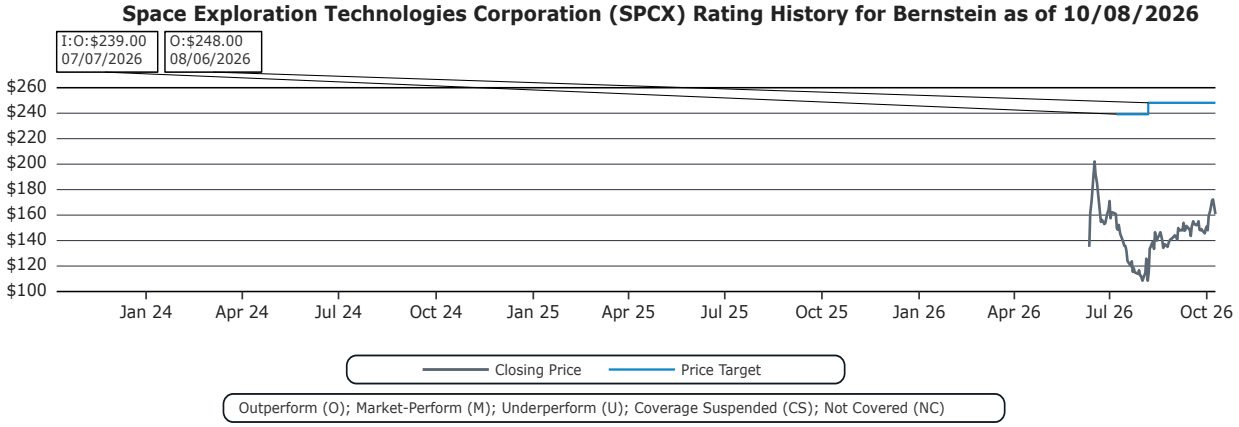
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	52.3%	14.2%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.6%	16.1%
Underperform	SELL	12.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of September 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

Prior to April 1, 2024, Sanford C. Bernstein & Co., LLC. issued the ratings and price target information in the graph(s) below for the following companies: Howmet Aerospace Inc.





All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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Stacy A. Rasgon maintains long positions in various crypto currencies.

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